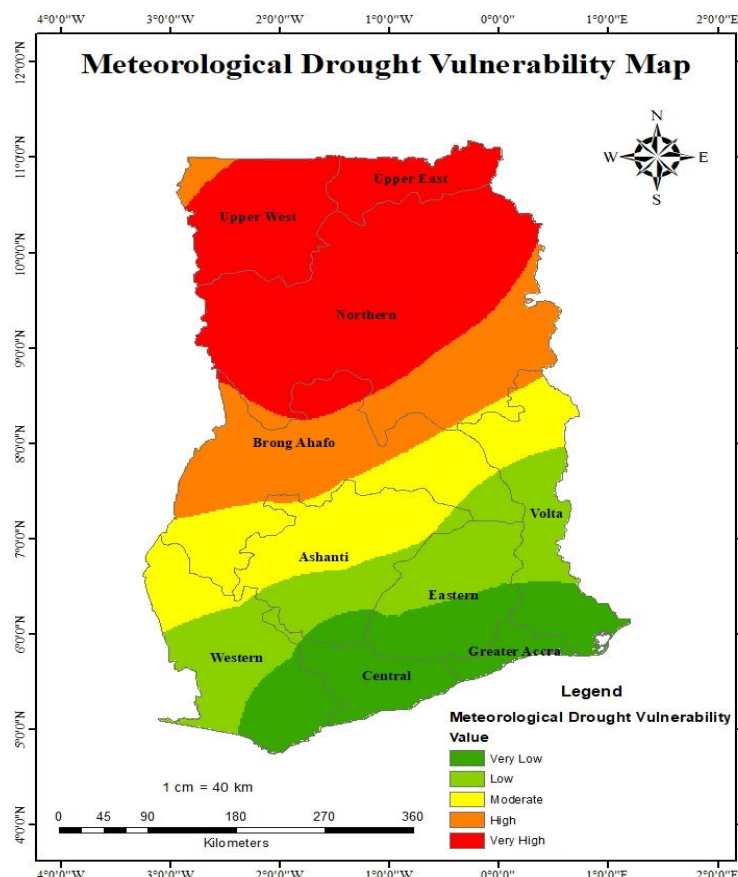


Figure 1:
Brought Map



Key Statistics at a Glance

- ✓ Target beneficiary: Smallholder farmers
- ✓ Estimated beneficiaries: 5 million people
- ✓ Jobs to be created: 10,000 direct and indirect employment
- ✓ Estimated Budget: 20 million US Dollars

Drought and The Looming Threat to Food Security: Policy Interventions for Northern Ghana

Executive Summary

The escalating frequency and intensity of droughts in Northern Ghana are jeopardizing food security, posing a significant threat to the livelihoods of millions of people. This policy brief outlines three possible interventions; investing in sustainable irrigation systems, promoting drought-resistant crops, and implementing a comprehensive climate change education program. The brief recommended a combination of these interventions be adopted to mitigate the drought's impact on food security. It also outlines the role of the different stakeholders and proposes a framework for the co-production of drought response measures.

Background

In Ghana, agriculture is an important contributor to the country's Gross Domestic Product (GDP) yet heavily dependent on rainfall which makes climate extremes like prolonged droughts a serious threat to food

security (Ndamani & Watanabe, 2014; Akudugu *et al.*, 2012). Northern Ghana happens to be an agricultural hub for most of the country's cereals, grains, legumes, roots and tubers, and livestock (Antwi-Agyei *et al.*, 2012). The region is, however, not exempt from extreme drought events, and this is exacerbated by its unimodal rainfall pattern (Ndamani and Watanabe, 2014). Figure 1 – 5 details measures of drought in Ghana.

Northern Ghana's agriculture-dependent economy is increasingly vulnerable due to the adverse effects of climate change, particularly drought. With over 70% of households engaged in agriculture, prolonged periods without rain significantly impact crop yields, leading to food shortages and increased poverty levels (Lu *et al.*, 2021).

Climatic and non-climatic stressors interact to affect agricultural practices and related livelihoods thereby forcing different adaptation measures such as intensification, growing drought-resistant crops, livelihood diversification, irrigation and growing of resistant varieties as asserted by Antwi-Agyei *et al.*, 2018 and Nyantakyi-Frimpong, 2013.

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Droughts do not only reduce yield but also the quality of the grain and hence the sales price, as pointed out by Gautier et al., 2016. The consequences of production losses include rises in food prices, food scarcity, and reduction in food consumption, with consequences on human health (including diseases and mortality, and childhood malnutrition affecting child growth and brain development). Tens of thousands of people died from starvation due to prolonged droughts and massive migrations took place during the 1983–1984 drought in the Sahel (Grote and Warner, 2010; De Bruijn & Van Dijk 1994). The situation is therefore urgent and demands immediate policy attention to avert a potential food security crisis.

Policy Options

The escalating impacts of drought on agriculture and food security demand urgent and effective interventions. As the situation continues to evolve, it becomes increasingly clear that we must explore a variety of adaptive strategies to ensure the sustainability and resilience of our agricultural systems. This section will present three potential policy interventions, each of these options with distinct advantages and challenges, which we will evaluate in terms of their potential benefits, drawbacks, and cost-effectiveness as shown in Table 1 in the appendix section. The ultimate goal is to identify robust strategies that can help our agricultural sector adapt to and thrive amidst the changing climate conditions. The policy options proposed therefore are:

1. **Investment in Sustainable Irrigation Systems:** By building and rehabilitating irrigation systems, the reliance on rainfall for crop cultivation could be reduced. This would also allow for year-round farming, increasing overall crop yields and could increase yield by 20 percent.
2. **Promotion of Drought-Resistant Crops:** Encourage the cultivation of drought-resistant and early-maturing crops like millets and cowpeas which can help reduce crop failure by 30 percent. This could involve subsidies, agricultural extension services, and improved access to quality seeds.
3. **Implement a Comprehensive Climate Change Education Program:** Increase public understanding of climate change and its impact on agriculture. Training farmers in innovative farming practices could enhance community resilience to drought and cut down losses by 50 percent.

Comparison of Policy Options

While all options are viable, they vary in cost, potential impact, and implementation timeframe. Irrigation systems provide a direct solution but are the most expensive and time-consuming to implement. Promoting drought-

resistant crops is less costly and can be implemented more quickly but relies on farmer buy-in. Climate change education is the least costly but requires a long-term commitment to realize its benefits.

Policy Recommendations

Given the critical nature of the threat posed by drought to food security in Northern Ghana, it is imperative that we act urgently and strategically to mitigate this issue. Based on our research and analysis, we strongly recommend the adoption of a three-pronged approach.

1. **Prioritize the Development of Sustainable Irrigation Systems:** As our first line of action, we must invest in the development and implementation of sustainable irrigation systems, especially in the most drought-prone areas. This investment will enhance the resilience of our farming systems, enabling them to maintain productivity despite unpredictable rainfall patterns. The associated costs are justified by the potential for improved agricultural yields, efficient water use, and long-term resilience of our food systems.
2. **Large-scale Promotion of Drought-Resistant Crops:** Concurrently, we propose a widespread campaign to encourage the cultivation of drought-resistant crops. This intervention will involve the provision of subsidies and quality seeds to farmers to facilitate the shift towards these more resilient crops. This strategy not only fosters adaptation to changing climate conditions but also helps reduce dependency on water resources, thus further reinforcing our food security.
3. **Implementation of a Comprehensive Climate Change Education Program:** Lastly, we must undertake a comprehensive climate change education program targeting farmers and other key stakeholders in the agricultural sector. By improving the understanding of climate change and its impacts on agriculture, we empower these key players to make informed decisions and develop adaptive strategies. The costs of this program are an investment in the long-term sustainability and resilience of our agricultural sector.

These recommended interventions are designed to work synergistically, providing a comprehensive approach to addressing the looming threat of drought to food security in Northern Ghana. Each component complements the others, resulting in a robust and holistic strategy.

Implementation Strategy

Effectively addressing the threat of drought to food security in Northern Ghana requires not only a robust policy framework, but also a carefully planned and executed implementation strategy. The following section outlines the key steps, stakeholders, and potential challenges associated with implementing the recommended policy interventions: the development of sustainable irrigation systems, the promotion of drought-resistant crops, and the implementation of a comprehensive climate change education program. Each of these strategies requires a unique approach, with a shared emphasis on collaboration, education, and sustainable funding. By detailing this strategic approach, we aim to provide a roadmap for turning these recommendations into tangible and effective actions. Refer to Table 2 for a detailed implementation strategy for this three-pronged approach solution.

Conclusion

As we navigate the escalating impacts of climate change, these interventions present a viable and strategic approach to enhancing the resilience of our agricultural sector and safeguarding our food security. It is our collective responsibility to act now, prioritizing sustainable and adaptive strategies that will ensure the long-term viability of agriculture and the well-being of our communities in Northern Ghana. While the road ahead may be challenging, the benefits of adopting these interventions far outweigh the costs. They offer a path

towards a more resilient and food-secure future for Northern Ghana amidst the challenges of climate change. It is our hope that this policy brief serves as a call to action for all stakeholders, inspiring urgent and coordinated efforts to address this critical issue.

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Appendices

Table 1: The Cost-Benefit Analysis of Proposed Policy Option for Northern Ghana

Policy Options	Pros	Cons	Cost-Benefit Analysis
1. Investment in Sustainable Irrigation Systems	- Increased agricultural productivity - Resource conservation	- Initial high cost - Training requirement	The benefits of improved productivity and resource conservation over time are likely to outweigh the initial costs and training requirements
2. Promotion of Drought-Resistant Crops	- Resilience to climate change - Reduced dependency on irrigation	- Potential for reduced crop diversity - Need for farmer education	The resilience of these crops and potential savings from reduced irrigation needs could outweigh the potential risks of reduced crop diversity and the cost of farmer education
3. Implementation of a Comprehensive Climate Change Education Program	- Increased awareness - Long-term adaptation strategies	- Time-consuming - Costs	The long-term benefits of improved understanding and adaptation to climate change may outweigh the time and cost of implementing the program

Table 2: Implementation Strategy for Policy Options

Policy Recommendations	Implementation Steps	Key Stakeholders	Potential Challenges and Mitigations
1. Development of Sustainable Irrigation Systems	a. Partner with organizations specializing in sustainable irrigation technology b. Identify drought-prone areas c. Secure funding d. Develop a farmer training program	Farmers, agricultural organizations, government agencies, private sector investors	Resistance to change (mitigate through effective communication and training), funding constraints (mitigate through strategic partnerships and diversified funding sources)
2. Promotion of Drought-Resistant Crops	a. Work with agricultural research institutions b. Develop a subsidy program c. Coordinate with seed suppliers d. Organize outreach programs	Farmers, agricultural research institutions, seed suppliers, government agencies	Possible lack of availability of quality seeds (mitigate through early coordination with suppliers), logistical issues (mitigate through careful planning)
3. Implementation of a Climate Change Education Program	a. Collaborate with educational institutions and environmental organizations b. Utilize various methods for program delivery c. Include follow-up mechanisms to assess effectiveness	Farmers, educational institutions, environmental organizations, agricultural extension officers	Resistance to change (mitigate through effective communication and training), funding constraints (mitigate through strategic partnerships and diversified funding sources)

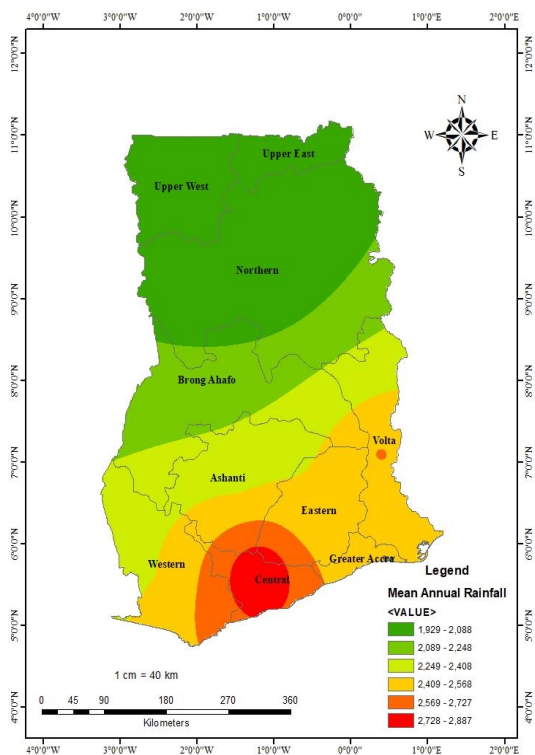


Figure 2: Mean Annual Rainfall for Ghana

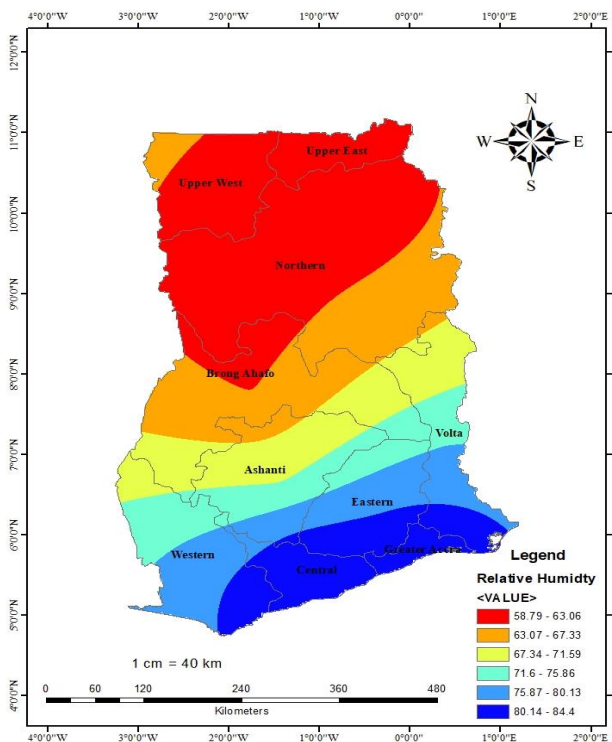


Figure 3: Relative Humidity Map for Ghana

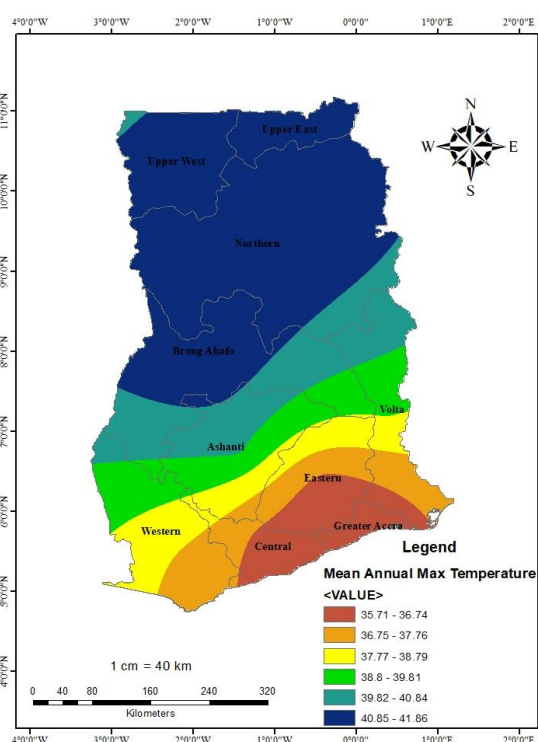


Figure 4: Mean Annual Maximum Temperature for Ghana

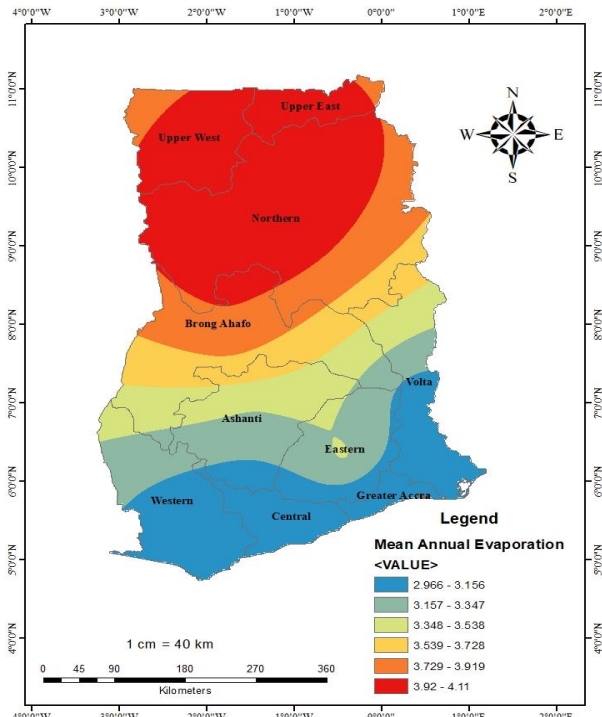


Figure 5: Mean Annual Evaporation for Ghana